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A discontinuous Galerkin framework for the radiative transfer approximation of high-frequency wave problems

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ABSTRACT

High-frequency wave energy transport is investigated in bounded domains with piecewise constant wave speed using a radiative transfer equation based model. We develop a powerful and efficient discontinuous Galerkin framework including both Fourier and discrete ordinate discretisations in the direction variable, which is widely applicable for wave problems arising within acoustics, elasticity and electromagnetics. Key features of the discontinuous Galerkin scheme implementation include fast exact quadrature for piecewise polynomial basis functions on general polygonal elements, together with an efficient parallel implementation. The scheme is benchmarked against both exact solutions and previous simulation results, demonstrating optimal convergence of the method with refinement of the grid as well as demonstrating convergence when increasing the spatial polynomial degree (alongside refinement of the directional discretisation) on a fixed coarse polygonal grid. As a final test, the scheme is applied to an example with more intricate geometric features requiring finer meshing and having millions of degrees of freedom. As well as producing converged results within competitive simulation times, the method was also used to investigate the inclusion of evanescent waves within the reflection and transmission laws for flexural plate motion between regions of different plate thickness. The expected trend was observed, with the inclusion of evanescent waves becoming less significant as the excitation frequency is increased.

1. Introduction

In the high-frequency limit, wave energy transport can be understood through the behaviour of the underlying ray dynamics in the regime where interference and other wave effects can be neglected. A wide class of high-frequency wave problems can thus be reduced to tracking ray densities that depend on both the position and a direction vector to specify the direction of travel. The differential equations governing these particle or ray densities are often expressed as conservation laws [1]; one example being the Liouville equation, which describes how ray densities evolve in Hamiltonian systems. This type of problem is especially relevant when approximating linear wave equations using their underlying ray dynamics in the short-wavelength limit. The radiative transfer equation (RTE) arises if we simplify the Liouville equation for propagation within (piecewise) homogeneous domains and include an absorption term to account for damping within the wave system [2]. Note that here we are not considering turbid media, and by RTE we refer to stationary radiative transfer models without in-scattering within the domain, as opposed to the RTE models often considered in biomedical imaging or atmospheric science applications [3–5]. The scattering-free stationary RTE is sometimes also termed the one-speed approximation of the steady-state first-order form of the neutron transport (FONT) equation for applications in nuclear reactor physics and radiation shielding problems [6,7].

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This study will focus on high-frequency wave problems arising in either an acoustics or vibration context, but we note that the framework is also directly applicable for applications in electromagnetics. Numerical approaches for solving ray density transport problems within acoustics and vibration have typically been based on boundary integral formulations using either the fundamental solution of the Liouville equation, which is the basis of Dynamical Energy Analysis (DEA) [8,9], or the fundamental solution of the stationary RTE [10,11]. In the latter case, the analysis was limited to diffuse reflections in order to make the computations more tractable, whereas DEA also models the directional dependence of the ray flow. Both approaches were put forward as more accurate and widely applicable than the popular Statistical Energy Analysis (SEA) method [12,13], which in addition to diffuse wave fields, requires a carefully chosen partition of the system into subsystems where the wave energy may be assumed to be in local equilibrium [14]. In fact, DEA was originally put forward as a method that both includes SEA as a low-order implementation and converges towards a ray-traced solution as the underlying basis approximation was extended to include more terms [8]. The advantage of these ray density transport formulations (including DEA) over more conventional ray tracing approaches [15,16] is that since they do not keep track of ray paths over multiple reflections, they do not suffer from potentially prohibitive computational costs when the number of these paths grows (sometimes exponentially) as the reflection order increases. In particular, ray-tracing can become inefficient in reverberant problems with chaotic ray-dynamics and in applications including ray splitting or scattering at boundaries [17,18]. A shortfall of DEA is that it is typically limited to first-order convergence when applied on domains with corners or edges, since the ray flow map appearing in the DEA boundary integral operator becomes discontinuous, hindering the convergence of numerical approximations [19].

In this work, instead of boundary integral formulations, we propose to use a discontinuous Galerkin (DG) approach for addressing ray density transport problems using the stationary RTE within an acoustics and vibration setting for the first time. For this initial study we will focus only on two-dimensional problems, with applications such as thin plate vibrations and room acoustics within horizontal cross-sections of acoustic cavities. Despite now discretising the full domain, we often obtain a piecewise-polynomial spatial discretisation of degree k with fewer degrees of freedom than DEA, since the application of DEA to two-dimensional meshed structures involves a discretisation over the element boundaries and the number of degrees of freedom per element is typically $k+1$ multiplied by the number of edges of the (polygonal) element [20]. The number of spatial degrees of freedom per element in our DG approach is $(k+1)(k+2)/2$, which is less than DEA for relatively small k values and the advantage is more significant for elements with more edges. It is also worth noting that DEA is most commonly applied with $k=0$; see [21] and references therein. A further advantage of a domain based discretisation is that it does not rely on a discontinuous boundary projected ray flow map, meaning that the convergence barrier experienced in DEA is no longer present.

The DG method was, in fact, originally developed for stationary transport equations [6] like the stationary RTE considered here, but in the setting of neutron transport. DG methods have also since been developed for the full RTE in turbid media with in-scattering [22,23]. We will develop our DG scheme for the scattering-free stationary RTE in complex domains with discontinuous wave-speeds; the DG approach has many advantages in this setting including the preservation of conserved quantities (mass, momentum, energy) [24], its inherent ability to handle solution discontinuities and complex geometries [25], and parallel scalability [26]. We can also draw upon a well-established body of theory and techniques; an excellent review article summarising many of these developments up to the turn of the century can be found in [27]. More recent developments that our study will build upon specifically include formulations with specular reflection boundary conditions [28,29], general polygonal meshes [30,31] and fast exact integration techniques for arbitrary degree piecewise polynomial bases [32,33].

We develop a DG spatial discretisation approach with fast exact integration methods on general polygonal meshes. The accompanying directional discretisation will be based on either a Fourier–Galerkin method or a more standard discrete ordinate (DO) approach. The Fourier approach may be advantageous in applications where an SEA approximation is valid (or almost valid), since if we truncate the Fourier expansion at three terms (with indices $-1, 0, 1$) and use only piecewise constant spatial basis functions then we obtain the SEA approximation when the mesh elements are taken as the SEA subsystems, providing a very efficient and small computational model. In general, the Fourier approximation with $2N+1$ terms corresponds to so-called the P_N approximation of the RTE [34,35], which we consider in the absence of in-scattering as discussed earlier. Alternatively, the DO method (or S_N approximation [7]) is ideal for applications where the wave or ray propagation is limited to a small number of known directions, as well as being the more standard approach used since the original DG formulation [6]. In comparison to the Fourier–Galerkin approach, DO is simpler and more efficient to implement since it does not require the inclusion of additional Fourier basis and test functions, nor the evaluation of a Galerkin projection integral over direction.

The remainder of this paper is structured as follows. In Section 2 we outline our stationary RTE model, including typical source terms, boundary conditions, interfacial behaviour in the presence of abrupt changes of wave speed and post-processing steps to compute the total wave energy density or fluence. Section 3 then details our DG spatial discretisation alongside either the Fourier or DO directional discretisation, discussing the weak form, fast exact quadrature and resulting linear system for both approaches alongside source terms, the implementation of reflection/transmission phenomena via an upwind numerical flux and the evaluation of the fluence. In Section 4, we present a series of numerical results of increasing complexity to verify the approach and showcase its versatility and capability. Finally, we present our conclusions in Section 5.

2. Problem formulation

We consider a bounded spatial domain $\Omega \subset \mathbb{R}^2$ representing either a linearly elastic structure or an acoustic medium. The objective is to characterise the distribution of high-frequency wave energy propagating through Ω using a reduced (scattering-free) form of the stationary RTE:

$$\hat{\boldsymbol{s}} \cdot \nabla u(\boldsymbol{x}, \hat{\boldsymbol{s}}) + \mu(\boldsymbol{x})\, u(\boldsymbol{x}, \hat{\boldsymbol{s}}) = f(\boldsymbol{x}, \hat{\boldsymbol{s}}), \qquad \boldsymbol{x} \in \Omega,\ \hat{\boldsymbol{s}} \in S^1. \tag{1}$$

Here $u(\boldsymbol{x}, \hat{\mathbf{s}})$ denotes the radiance, representing the directional energy density at position $\boldsymbol{x}$ travelling in direction $\hat{\mathbf{s}}(\theta) = (\cos\theta, \sin\theta)$ on the unit circle S^1. The coefficient $\mu(\boldsymbol{x})$ describes attenuation or damping within Ω, while $f(\boldsymbol{x}, \hat{\mathbf{s}})$ represents a prescribed source function or distribution.

We impose specular reflection conditions at the boundary $\Gamma = \partial\Omega$. Let $\hat{\boldsymbol{n}}(\boldsymbol{x})$ denote the outward unit normal at $\boldsymbol{x} \in \Gamma$ and $u_0(\boldsymbol{x}, \hat{\mathbf{s}})$ represent a prescribed boundary source that injects energy into the domain, with $u_0(\boldsymbol{x}, \hat{\mathbf{s}}) = 0$ on source-free regions of Γ. A specular reflection corresponds to a perfectly reflecting wall with no energy absorption or scattering [36] and the corresponding boundary condition is given by

$$u(\boldsymbol{x}, \hat{\mathbf{s}}) = u_0(\boldsymbol{x}, \hat{\mathbf{s}}) + u\left(\boldsymbol{x},\ \hat{\mathbf{s}} - 2\left(\hat{\mathbf{s}} \cdot \hat{\boldsymbol{n}}\right)\hat{\boldsymbol{n}}\right), \quad \boldsymbol{x} \in \Gamma, \quad \hat{\mathbf{s}} \cdot \hat{\boldsymbol{n}} < 0. \tag{2}$$

The right side of Eq. (2) therefore corresponds to the sum of a source term and reflection term (corresponding to rays reflecting into direction $\hat{\mathbf{s}}$ at $\boldsymbol{x} \in \Gamma$), and the condition $\hat{\mathbf{s}} \cdot \hat{\boldsymbol{n}} < 0$ restricts to directions propagating into Ω at $\boldsymbol{x} \in \Gamma$.

The fluence Φ provides a direction-integrated measure of the total wave energy density at each point in the domain. It is defined in terms of $u(\boldsymbol{x}, \hat{\mathbf{s}})$ as

$$\Phi(\boldsymbol{x}) = \int_{S^1} u(\boldsymbol{x}, \hat{\mathbf{s}})\, \mathrm{d}\hat{\mathbf{s}} = \int_{-\pi}^{\pi} u(\boldsymbol{x}, \hat{\mathbf{s}}(\theta))\, \mathrm{d}\theta, \tag{3}$$

and represents the total energy density obtained by summing contributions from all propagation directions.

Energy may be introduced into the system through boundary and/or internal sources. Boundary sources are incorporated via the $u_0(\boldsymbol{x}, \hat{\mathbf{s}})$ term appearing in Eq. (2). Internal sources are incorporated directly in the right-hand side of Eq. (1) via $f(\boldsymbol{x}, \hat{\mathbf{s}})$. An omnidirectional point source located at $\boldsymbol{x}_0 \in \Omega$ may be included by taking

$$f(\boldsymbol{x}, \hat{\mathbf{s}}) = \frac{1}{2\pi}\delta(\boldsymbol{x} - \boldsymbol{x}_0), \tag{4}$$

where δ denotes the Dirac delta. In practice, the singular nature of the Dirac delta in Eq. (4) cannot be represented accurately using the spatial polynomial basis of the discontinuous Galerkin formulation, which we later use to discretise Eq. (1) in space. To overcome this, once discretised, the point source is treated by projecting it onto the boundary of the element that contains $\boldsymbol{x}_0$. Therefore, the right hand side of (1) is set to zero and the problem is reformulated for a boundary-source along the local element boundary (reflected contribution) and its directly neighbouring elements (transmitted contribution) similar to the incorporation of boundary sources described above. Within the source-containing element, the local fluence is supplemented with the free-space Green's function for the RTE, providing an exact representation of the direct source contribution (before any reflections) in the vicinity of the source.

We will consider domains Ω containing different acoustic media or different material properties, each characterised by distinct local parameters (density, wave speed, attenuation, thickness etc.). These variations are represented by spatially piecewise constant parameters. Discontinuities in wave speed/impedance at interfaces between these different media/materials will give rise to reflection and transmission described by the Fresnel equations and governed by continuity relations imposed across the interface [20,37]. The direction of the transmitted wave components may be obtained from Snell's law,

$$\eta_I \sin\theta_I = \eta_T \sin\theta_T, \tag{5}$$

where $\eta = 1/c$ denotes the slowness of the medium and the subscripts I and T refer to the incident and transmitted waves, respectively. Together with the Fresnel equations, this relation defines the interfacial behaviour governing energy transport across abrupt changes of medium/material.

3. Discretisation of the radiative transfer equation

In this section, we present the discretisation framework adopted for the numerical solution of the RTE, capturing both spatial and directional dependencies using separate methodologies. The spatial discretisation is performed using the DG method, while the directional discretisation is addressed through either a discrete ordinates approach or a Fourier–Galerkin formulation.

3.1. Discretisation methods and weak formulations

To model the spatial dependence we apply the DG method to (1) by initially partitioning the spatial domain Ω into N_h non-overlapping polygonal elements E_j, with $j = 1, 2, \ldots, N_h$. Each polygonal element E_j has κ_j edges, denoted $\Gamma_{i,j}$ for $i = 1, 2, \ldots, \kappa_j$. To represent the local behaviour of the solution within each element, we define an element-wise approximation $u_j(\boldsymbol{x}, \theta) \approx u(\boldsymbol{x}, \hat{\mathbf{s}})$ for $\boldsymbol{x} \in E_j$, and set $u_j(\boldsymbol{x}, \theta) = 0$ otherwise, for $j = 1, 2, \ldots, N_h$. We also assume that the attenuation coefficient $\mu(\boldsymbol{x})$ is locally constant within each element and hence set $\mu(\boldsymbol{x}) = \mu_j$ for $\boldsymbol{x} \in E_j$, where μ_j, $j = 1, \ldots, N_h$ is a set of known constant values. The spatial dependence of u_j is approximated using scaled Legendre polynomials of degree $n = 0, 1, 2, \ldots$, defined by

$$\tilde{P}_n(\xi) = \sqrt{\frac{2n+1}{2}}\, P_n(\xi), \quad -1 \le \xi \le 1, \tag{6}$$

where P_n denotes the standard Legendre polynomial of degree n. These scaled polynomials are orthonormal, satisfying

$$\int_{-1}^{1} \tilde{P}_m(\xi)\, \tilde{P}_n(\xi)\, \mathrm{d}\xi = \delta_{mn},$$

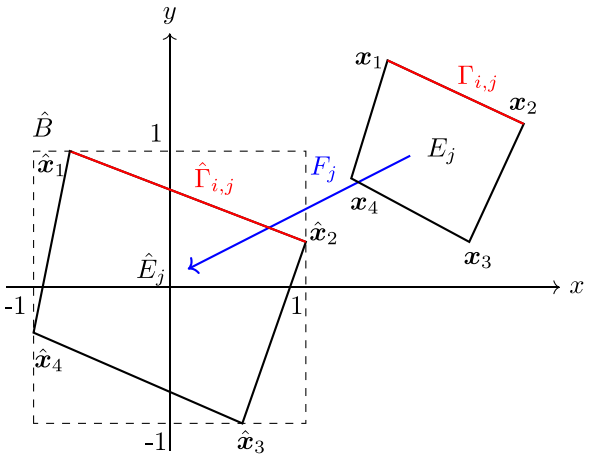


Fig. 1. Visualisation of the affine map F_j, defined in (7), when applied to a quadrilateral element E_j in order to obtain the mapped element $\hat{E}_j \subset \hat{B} = (-1,1)^2$. A quadrilateral element is used for illustration purposes only; the same mapping procedure is applied for any polygonal element geometry.

where δ_{mn} represents the Kronecker delta.

To implement (6) as DG basis functions over general polygonal elements in conjunction with the fast exact quadrature introduced in [32], each element $E_j \subset \Omega$ is enclosed within its minimal Cartesian bounding box

$$B_{E_j} = \left(x_c - \frac{h_x}{2}, x_c + \frac{h_x}{2}\right) \times \left(y_c - \frac{h_y}{2}, y_c + \frac{h_y}{2}\right)$$

with side lengths h_x and h_y, and centroid (x_c, y_c). We then introduce the following affine mapping,

$$(\hat{x}, \hat{y}) = \hat{\boldsymbol{x}} = F_j(\boldsymbol{x}) := \left(\frac{2\left(x - x_c\right)}{h_x}, \frac{2\left(y - y_c\right)}{h_y}\right), \tag{7}$$

to map $(x, y) = \boldsymbol{x} \in E_j \subset B_{E_j}$ onto $\hat{\boldsymbol{x}} \in \hat{E}_j$ within the reference box $\hat{B} = (-1,1)^2$ as shown in Fig. 1. The κ_j edges of the mapped polygon $\hat{E}_j$ are correspondingly denoted $\hat{\Gamma}_{i,j}$ for $i = 1, 2, \ldots, \kappa_j$. The Jacobian of the inverse transformation F_j^{-1} is $|\boldsymbol{J}_E| = \frac{1}{4} h_x h_y$, and along an edge $\Gamma_{i,j} \subset E_j$, the corresponding Jacobian is given by $|\boldsymbol{J}_\Gamma| = \|\boldsymbol{J}_E^\top \hat{\boldsymbol{n}}\| \, |\boldsymbol{J}_E|$, where the norm notation refers to the Euclidean length [32]. The transformation (7) will prove useful for the exact evaluation of the integrals arising in the DG weak formulation for convex polygonal elements using the integration framework introduced in [32].

For the directional discretisation we employ either a Fourier-based approach or the DO method. The DO method is particularly effective for cases where the energy propagates in a finite number of discrete directions, whereas the Fourier-based formulation is advantageous for problems with a smoother directional dependence.

In the Fourier-based approach, a directional Fourier expansion is combined with the spatial Legendre polynomial basis $\tilde{P}_{\alpha_1}(\hat{x})\,\tilde{P}_{\alpha_2}(\hat{y})$ from (6), where α_1 and α_2 represent the polynomial degrees in $\hat{x}$ and $\hat{y}$ respectively, satisfying $\alpha_1 + \alpha_2 \leq k$. We note that spatial basis functions of this form are a popular alternative to standard finite element nodal basis functions in DG [27], having several advantages including direct applicability to different element geometries and the simplicity of (p-)refinement of the scheme by increasing k [32], as well as optimal convergence results in certain settings [38,39]. The fast exact quadrature introduced in [32] is also more conveniently applied when $\alpha_1 + \alpha_2 \leq k$. In fact, it is most simply applied for monomial (or so-called Pascal's triangle [29]) basis functions, but can easily be adapted for other polynomial bases such as the Legendre polynomials employed here [32]. The local approximation $u_j(\hat{\boldsymbol{x}}, \theta)$ is then written as

$$u_j(\hat{\boldsymbol{x}}, \theta) = \sum_{\alpha_1=0}^{k} \sum_{\alpha_2=0}^{k-\alpha_1} \sum_{n=-N}^{N} u^n_{j,\alpha_1,\alpha_2} \tilde{P}_{\alpha_1}(\hat{x})\,\tilde{P}_{\alpha_2}(\hat{y}) \frac{e^{in\theta}}{\sqrt{2\pi}}. \tag{8}$$

The weak formulation is obtained by multiplying by a test function $v(\hat{\boldsymbol{x}}, \theta)$ drawn from the same finite-dimensional space as the basis functions in (8), and integrating over E_j and θ. In this setting, the test function is chosen as

$$v_j(\hat{\boldsymbol{x}}, \theta) = \tilde{P}_{\beta_1}(\hat{x})\,\tilde{P}_{\beta_2}(\hat{y}) \frac{e^{-im\theta}}{\sqrt{2\pi}}, \tag{9}$$

where $m = -N, -N+1, \ldots, N$ and (β_1, β_2) take the same index values as (α_1, α_2) in (8).

Alternatively, for the DO method, the directional dependence of $u_j(\hat{\boldsymbol{x}},\theta)$ is evaluated at a set of N_θ uniformly spaced directions θ_n for $n = 1, 2, \dots, N_\theta$. For each θ_n and $\boldsymbol{x} \in E_j$, we write

$$u_j(\hat{\boldsymbol{x}},\theta_n) = u_j^n(\hat{\boldsymbol{x}}) = \sum_{\alpha_1=0}^{k}\sum_{\alpha_2=0}^{k-\alpha_1} u^n_{j,\alpha_1,\alpha_2}\tilde{P}_{\alpha_1}(\hat{x})\,\tilde{P}_{\alpha_2}(\hat{y}), \quad \alpha_1+\alpha_2 \le k, \quad n = 1,2,\dots,N_\theta, \tag{10}$$

and the test functions are now taken as

$$w_j(\hat{\boldsymbol{x}}) = \tilde{P}_{\beta_1}(\hat{x})\,\tilde{P}_{\beta_2}(\hat{y}),$$

for $\beta_1 + \beta_2 \le k$.

For the Fourier method, we find the local weak form by substituting the approximation (8) into the RTE (1), multiplying by v_j in (9), and integrating over both $\boldsymbol{x} \in E_j$ and $\theta \in [-\pi,\pi)$ to give

$$\int_{-\pi}^{\pi}\int_{\hat{E}_j}\left(\hat{\mathbf{s}}(\theta)\cdot\nabla u_j(\hat{\boldsymbol{x}},\theta) + \mu_j u_j(\hat{\boldsymbol{x}},\theta)\right) v_j(\hat{\boldsymbol{x}},\theta)\,|\boldsymbol{J}_{E_j}|\,\mathrm{d}\hat{\boldsymbol{x}}\,\mathrm{d}\theta = 0. \tag{11}$$

Note that here we set $f \equiv 0$ since the source terms for both boundary and point sources enter via a boundary contribution as detailed in Section 2. Note also that we have made use of the map F_j introduced in (7) to change the spatial integration variable to $\hat{\boldsymbol{x}} \in \hat{E}_j \subset (-1,1)^2$. We obtain the aforementioned boundary contribution by applying Green's first identity, which yields

$$\begin{aligned}&\int_{-\pi}^{\pi}\int_{\hat{E}_j}\mu_j u_j(\hat{\boldsymbol{x}},\theta)\,v_j(\hat{\boldsymbol{x}},\theta)\,|\boldsymbol{J}_{E_j}|\,\mathrm{d}\hat{\boldsymbol{x}}\,\mathrm{d}\theta - \int_{-\pi}^{\pi}\int_{\hat{E}_j} u_j(\hat{\boldsymbol{x}},\theta)\,(\hat{\mathbf{s}}(\theta)\cdot\nabla v_j(\hat{\boldsymbol{x}},\theta))|\boldsymbol{J}_{E_j}|\,\mathrm{d}\hat{\boldsymbol{x}}\,\mathrm{d}\theta\\ &+\sum_{i=1}^{\kappa_j}\int_{-\pi}^{\pi}\int_{\hat{\Gamma}_{i,j}}\gamma_{i,j}(\hat{\boldsymbol{x}},\theta)\,v_j(\hat{\boldsymbol{x}},\theta)\,|\boldsymbol{J}_{\Gamma_{i,j}}|\,\mathrm{d}\hat{\boldsymbol{x}}\,\mathrm{d}\theta = 0.\end{aligned} \tag{12}$$

We assume that the underlying properties of the media/material represented by Ω are piecewise constant, in the same way as the attenuation coefficient outlined earlier, and hence the wave speed $c > 0$ may be expressed as $c(\boldsymbol{x}) = c_j$, when $\boldsymbol{x} \in E_j$ with c_j, $j = 1,\dots,N_h$ a set of known constants. On each edge $\Gamma_{i,j} \subset E_j$ with outward normal $\hat{\boldsymbol{n}}(\boldsymbol{x})$, $\gamma_{i,j}(\hat{\boldsymbol{x}},\theta)$ defines an upwind scheme incorporating Fresnel reflection/transmission and the boundary condition (2) as follows:

$$\gamma_{i,j}(\hat{\boldsymbol{x}},\theta) = \begin{cases}(\hat{\mathbf{s}}(\theta)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}))\,u_j(\hat{\boldsymbol{x}},\theta) + \left(\hat{\mathbf{s}}\left(\theta_R\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right)\mathcal{R}_{j,j_+}(\theta)\,u_j\left(\hat{\boldsymbol{x}},\theta_R\right), & \hat{\mathbf{s}}(\theta)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}) > 0,\\ (\hat{\mathbf{s}}(\theta)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}))\,\mathcal{T}_{j_+,j}(\theta_{T_+})\,u_{j_+}\left(\hat{\boldsymbol{x}},\theta_{T_+}\right), & \hat{\mathbf{s}}(\theta)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}) < 0,\ \boldsymbol{x}\notin\Gamma,\\ (\hat{\mathbf{s}}(\theta)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}))\,u_0\left(\boldsymbol{x},\hat{\mathbf{s}}(\theta)\right), & \hat{\mathbf{s}}(\theta)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}) < 0,\ \boldsymbol{x}\in\Gamma,\end{cases} \tag{13}$$

noting that $\boldsymbol{x} = F_j^{-1}(\hat{\boldsymbol{x}})$. Here, u_{j+} denotes the local DG approximation within the neighbouring element E_{j_+} that lies upwind of E_j, connected through $\Gamma_{i,j} = \Gamma_{i+,j+} \subset E_{j+}$ [24]. At any point $\boldsymbol{x} \in \Gamma_{i,j}$, the specularly reflected direction in E_j is

$$\hat{\mathbf{s}}(\theta_R) = \hat{\mathbf{s}}(\theta) - 2\left(\hat{\mathbf{s}}(\theta)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right)\hat{\boldsymbol{n}}(\boldsymbol{x}).$$

We also let θ_{T_+} denote the incident direction in E_{j_+} that refracts to the direction θ in E_j. As a result, the notation $\mathcal{T}_{j_+,j}(\theta_{T_+})$ indicates the Fresnel transmission factor applied to the neighbour field u_{j_+} as it passes from E_{j_+} into E_j across $\Gamma_{i+,j+}$. Likewise, $\mathcal{R}_{j,j_+}$ indicates rays reflecting into E_j, rather than transmitting into E_{j_+}. For interfaces within homogeneous regions of Ω we set $\mathcal{R}_{j,j_+} = 0, \mathcal{T}_{j_+,j} = 1$. Conversely, for perfectly reflective boundaries at $\Gamma_{i,j} \subset \Gamma$, the reflective part of the boundary condition in Eq. (2) is enforced by setting $\mathcal{R}_{j,0} = 1$, where the $'0'$ subscript represents the absence of a neighbouring element on the other side of Γ.

For the discrete ordinates discretisation, the procedure is similar to the one above except for evaluating at a set of discrete θ values, $\theta = \theta_n$, $n = 1,\dots,N_\theta$, instead of imposing a weak form in θ. We obtain

$$\begin{aligned}&\int_{\hat{E}_j}\mu_j u_j^n(\hat{\boldsymbol{x}})\,w_j(\hat{\boldsymbol{x}})\,|\boldsymbol{J}_{E_j}|\,\mathrm{d}\hat{\boldsymbol{x}} - \int_{\hat{E}_j} u_j^n(\hat{\boldsymbol{x}})\,(\hat{\mathbf{s}}(\theta_n)\cdot\nabla w_j(\hat{\boldsymbol{x}}))|\boldsymbol{J}_{E_j}|\,\mathrm{d}\hat{\boldsymbol{x}}\\ &+\sum_{i=1}^{\kappa_j}\int_{\hat{\Gamma}_{i,j}}\gamma_{i,j}\left(\hat{\boldsymbol{x}},\theta_n\right)w_j(\hat{\boldsymbol{x}})\,|\boldsymbol{J}_{\Gamma_{i,j}}|\,\mathrm{d}\hat{\boldsymbol{x}} = 0,\end{aligned} \tag{14}$$

for $n = 1,2,\dots,N_\theta$ and $\gamma_{i,j}(\hat{\boldsymbol{x}},\theta_n)$ is given by

$$\gamma_{i,j}(\hat{\boldsymbol{x}},\theta_n) = \begin{cases}\left(\hat{\mathbf{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right)u_j^n(\hat{\boldsymbol{x}}) + \left(\hat{\mathbf{s}}\left(\theta_R\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right)\mathcal{R}_{j,j_+}(\theta_n)\,u_j\left(\hat{\boldsymbol{x}},\theta_R\right), & \hat{\mathbf{s}}(\theta_n)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}) > 0,\\ \left(\hat{\mathbf{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right)\mathcal{T}_{j_+,j}(\theta_{T_+})\,u_{j_+}\left(\hat{\boldsymbol{x}},\theta_{T_+}\right), & \hat{\mathbf{s}}(\theta_n)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}) < 0,\ \boldsymbol{x}\notin\Gamma,\\ \left(\hat{\mathbf{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right)u_0\left(\boldsymbol{x},\hat{\mathbf{s}}(\theta_n)\right), & \hat{\mathbf{s}}(\theta_n)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}) < 0,\ \boldsymbol{x}\in\Gamma.\end{cases} \tag{15}$$

In the DO method, the reflected direction θ_R and the pre-transmitted direction θ_{T_+} may not be exactly contained within the discrete ordinates sample θ_n, $n = 1,\dots,N_\theta$. We therefore let k_1, k_2 be the indices of the two ordinates that are closest to θ_* ($*$ represents either R or T_+ here) and obtain the value of $u_j\left(\hat{\boldsymbol{x}},\theta_*\right)$ by linear interpolation from the values $u_j^{k_1}(\hat{\boldsymbol{x}})$ and $u_j^{k_2}(\hat{\boldsymbol{x}})$. That is, we set

$$u_j\left(\hat{\boldsymbol{x}},\theta_*\right) = \frac{1}{|\theta_{k_1}-\theta_{k_2}|}\left(|\theta_*-\theta_{k_2}|u_j^{k_1}(\hat{\boldsymbol{x}}) + |\theta_{k_1}-\theta_*|u_j^{k_2}(\hat{\boldsymbol{x}})\right).$$

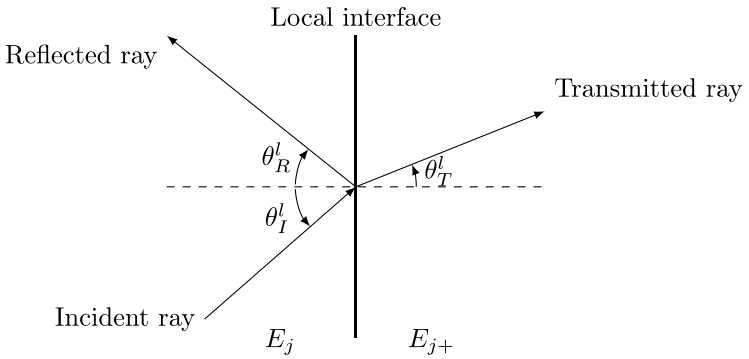


Fig. 2. Reflection/transmission at the internal element boundary between the element E_j and its neighbouring element E_{j+}.

We note that a similar interpolation scheme was proposed for implementing specular reflections within a DO-based DG scheme in Ref. [28].

3.2. Reflection and transmission at internal element boundaries

The reflection and transmission factors are computed by mapping the global angle $\theta \in [-\pi, \pi)$ to a local incident angle $\theta_I^l \in (-\pi/2, \pi/2)$ relative to the normal vector pointing into the ray-containing element. We then obtain θ_R^l and θ_T^l via Snell's Law and use these to calculate the reflection/transmission coefficients, before finally transforming these angles back to the global direction coordinate system. We consider an incident ray within E_j travelling towards an interface with the neighbouring element E_{j+}, having incidence angle θ_I^l. Upon arrival at the interface, this results in a reflected ray travelling away from the interface with local angle θ_R^l, and a transmitted ray travelling in the neighbouring element E_{j+} with local direction θ_T^l. See Fig. 2 for a depiction of this set up. The transmitted angle in local form is determined by Snell's Law (5) via

$$\theta_T^l = \sin^{-1}\left(\frac{c_{j+}}{c_j}\sin\theta_I^l\right),$$

where the case $|(\eta_I/\eta_T)\sin\theta_I^l| > 1$ corresponds to an evanescent transmitted field.

For flexural plate waves, the reflection/transmission coefficients are obtained by enforcing continuity of displacement, bending moment, effective shear, and twisting moment across the interface [40]. This yields a 4×4 linear system for the complex amplitudes of the reflected/transmitted propagating modes together with the two evanescent components. In the high-frequency regime, the evanescent contributions decay over very short distances and can be neglected. The interface problem then reduces to a 2×2 system and the reflection/transmission coefficients arising from an incident wave in E_j arriving at the interface with E_{j+} become

$$\mathcal{R}_{j,j_+}(\theta_I) = \left(\frac{k_j\cos\theta_I^l - k_{j+}\cos\theta_T^l}{k_j\cos\theta_I^l + k_{j+}\cos\theta_T^l}\right)^2, \qquad \mathcal{T}_{j,j_+}(\theta_I) = \frac{4\,k_j k_{j+}\cos\theta_I^l\cos\theta_T^l}{\left(k_j\cos\theta_I^l + k_{j+}\cos\theta_T^l\right)^2}. \tag{16}$$

Here, the wavenumbers are defined by $k_j = \omega/c_j$ and $k_{j+} = \omega/cj+$, with ω the angular frequency associated with the source term u_0.

For acoustic plane waves in fluids, the natural material parameter is the acoustic impedance $z_j = \rho_j c_j$, where ρ_j is the local fluid density in E_j and c_j is the local wave speed, as before. The interface continuity conditions are instead taken as the continuity of acoustic pressure and the particle velocity in the direction normal to the interface. For the same general interface between E_j and E_{j+}, with impedances of z_j and z_{j+}, respectively, the reflection and transmission factors are a simple modification of Eq. (16) by interchanging k_j with z_{j+} and k_{j+} with z_j.

3.3. Source terms

In the case where a boundary source is present along the ith edge $\Gamma_{i,j}$ of E_j with $\Gamma_{i,j} \subset \Gamma$, then this leads to a non-zero contribution for u_0 in Eq. (2). For example, a directional boundary source propagating in the fixed direction θ_0 (into Ω) is obtained by setting $u_0(\boldsymbol{x}, \hat{\boldsymbol{s}}(\theta)) = \delta(\theta - \theta_0)$ with $\hat{\boldsymbol{s}}(\theta_0)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}) < 0$. Typically, we apply this type of source on a small region of Γ, rather than on the entire boundary. After discretisation, the source contribution in element E_j is expressed as $u_{0_j}(\boldsymbol{x}, \theta) \approx u_0(\boldsymbol{x}, \hat{\boldsymbol{s}})$ for $\boldsymbol{x} \in E_j$. This local source contribution is then projected onto the DG-Fourier basis by expanding in a sum of the form

$$u_{0_j}(\hat{\boldsymbol{x}}, \theta) = \sum_{\beta_1=0}^{k}\sum_{\beta_2=0}^{k-\beta_1}\sum_{m=-N}^{N} u^m_{0_{j,\beta_1,\beta_2}}\tilde{P}_{\beta_1}(\hat{x})\,\tilde{P}_{\beta_2}(\hat{y})\frac{e^{\mathrm{i}m\theta}}{\sqrt{2\pi}}. \tag{17}$$

The expansion coefficients $u^m_{0_{j,\beta_1,\beta_2}}$ for the DG-Fourier source expansion (17) of $u_0(\boldsymbol{x},\hat{\boldsymbol{s}}) = \delta\big(\theta-\theta_0\big)$ may then be determined along edges $\Gamma_{i,j} \subset \Gamma$ of E_j by evaluating the final expression on the left of Eq. (12). We obtain

$$u^m_{0_{j,\beta_1,\beta_2}} = -\sum_{\Gamma_{i,j}\subset\Gamma}\int_{-\pi}^{\pi}\int_{\hat{\Gamma}_{i,j}} \left(\hat{\boldsymbol{s}}\left(\theta\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right) v_j\left(\hat{\boldsymbol{x}},\theta\right)\delta\big(\theta-\theta_0\big)|\boldsymbol{J}_{\Gamma_{i,j}}|\,\mathrm{d}\hat{\boldsymbol{x}}\,\mathrm{d}\theta,$$

where the minus sign arises from moving this known expression onto the right side of (12). We can then simplify using the sifting property of the Dirac delta to yield

$$u^m_{0_{j,\beta_1,\beta_2}} = -\sum_{\Gamma_{i,j}\subset\Gamma}\int_{\hat{\Gamma}_{i,j}} \left(\hat{\boldsymbol{s}}\left(\theta_0\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right) v_j\left(\hat{\boldsymbol{x}},\theta_0\right)|\boldsymbol{J}_{\Gamma_{i,j}}|\,\mathrm{d}\hat{\boldsymbol{x}}.$$

For the DO method, we choose the ordinates to ensure that $\theta_0 \in \{\theta_n\}_{n=1,\ldots,N_\theta}$ and expand $u_{0_j}(\boldsymbol{x},\theta)$ for each $\theta=\theta_n$, $n=1,\ldots,N_\theta$ as

$$u_{0_j}(\hat{\boldsymbol{x}},\theta_n) = \sum_{\beta_1=0}^{k}\sum_{\beta_2=0}^{k-\beta_1} u^n_{0_{j,\beta_1,\beta_2}}\tilde{P}_{\beta_1}(\hat{x})\,\tilde{P}_{\beta_2}(\hat{y}). \tag{18}$$

The expansion coefficients $u^n_{0_{j,\beta_1,\beta_2}}$ for the DG-DO source expansion (18) of $u_0(\boldsymbol{x},\hat{\boldsymbol{s}}) = \delta\big(\theta-\theta_0\big)$ are then calculated by evaluating the final expression on the left of Eq. (14). We obtain

$$u^n_{0_{j,\beta_1,\beta_2}} = -\sum_{\Gamma_{i,j}\subset\Gamma}\int_{\hat{\Gamma}_{i,j}} \left(\hat{\boldsymbol{s}}\left(\theta_0\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right) w_j\left(\hat{\boldsymbol{x}}\right)|\boldsymbol{J}_{\Gamma_{i,j}}|d\hat{\boldsymbol{x}}, \tag{19}$$

if $\theta_0=\theta_n$, or $u^n_{0_{j,\beta_1,\beta_2}} = 0$ otherwise. As before, the minus sign arises from moving the known source expression onto the right side of (14).

For an internal point source of the form (4), then as described in Section 2 we project the source contribution onto the boundaries of the element E_p, which denotes the element containing the source point $\boldsymbol{x}_0$. The direct contribution (before any reflections) from the point source to the boundary of E_p can be described using the free-space Green's function for (1) via

$$u_0(\boldsymbol{x},\hat{\boldsymbol{s}}) = \frac{e^{-\mu_p\|\boldsymbol{x}-\boldsymbol{x}_0\|}}{2\pi\|\boldsymbol{x}-\boldsymbol{x}_0\|}\delta\big(\theta-\theta_0(\boldsymbol{x})\big), \tag{20}$$

where μ_p is the local attenuation coefficient within E_p and the norm notation denotes the Euclidean distance from the source point $\boldsymbol{x}_0$ to the point $\boldsymbol{x}$ on the boundary of E_p. The angle $\theta_0(\boldsymbol{x})$ is associated with the direction vector

$$\hat{\boldsymbol{s}}_0 = (\cos(\theta_0(\boldsymbol{x})),\sin(\theta_0(\boldsymbol{x}))) = \left(\frac{x-x_0}{\|\boldsymbol{x}-\boldsymbol{x}_0\|},\frac{y-y_0}{\|\boldsymbol{x}-\boldsymbol{x}_0\|}\right),$$

pointing from $\boldsymbol{x}_0=(x_0,y_0)$ to $\boldsymbol{x}=(x,y)$ on the boundary of E_p. Note that the Green's function (20) is proportional to the square modulus of the leading-order high-frequency asymptotic formula for the free-space Green's function for both the Helmholtz equation and the biharmonic Helmholtz equation (for flexural waves in plates) after including damping and directional dependence [20], and so all that is required to apply this point source model to different wave equations is to rescale by a constant.

In the Fourier method, the boundary flux basis coefficients for a source point located in E_p may, in general, give rise to a reflected contribution from each edge of E_p. The coefficients of the source expansion (17) for this reflected contribution are given by substituting the known source term (20) for u_j in the reflected contribution within (13) and then substituting the result into the final term on the left of (12). We obtain

$$u^m_{0_{p,\beta_1,\beta_2}} = \sum_{i=1}^{\kappa_p}\int_{\hat{\Gamma}_{i,p}} \left(\hat{\boldsymbol{s}}\left(\theta_0(\boldsymbol{x})\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right) v_p\left(\hat{\boldsymbol{x}},\theta_0^R(\boldsymbol{x})\right)\mathcal{R}_{p,p_+}(\theta_0(\boldsymbol{x}))\frac{e^{-\mu_p\|\boldsymbol{x}-\boldsymbol{x}_0\|}}{2\pi\|\boldsymbol{x}-\boldsymbol{x}_0\|}|\boldsymbol{J}_{\Gamma_{i,p}}|\,\mathrm{d}\hat{\boldsymbol{x}}, \tag{21}$$

where θ_0^R is the resulting direction after the source ray with direction θ_0 is specularly reflected along one of the edges of E_p and we have used that $\hat{\boldsymbol{s}}\left(\theta_0\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}) = -\hat{\boldsymbol{s}}\left(\theta_0^R\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})$ to incorporate the minus sign given by moving the known source term to the right hand side. We note that the Dirac delta in (20) has been used to evaluate the integral over θ using the sifting property. The contribution from (21) will be zero if E_p is in a homogeneous region of Ω where the properties of the neighbouring elements all match those of E_p. We also have a transmitted source contribution along the adjoining edge $\Gamma_{i_p,p+}$ of any neighbouring element E_{p+} to E_p, whose expansion coefficients are given by substituting the known source term (20) for u_j into the transmitted contribution in (13) and then substituting the result into the final term on the left of (12). We obtain

$$\begin{aligned} u^m_{0_{p+,\beta_1,\beta_2}} &= \\ &-\int_{\hat{\Gamma}_{i_p,p+}} \left(\hat{\boldsymbol{s}}\left(\theta_0^T(\boldsymbol{x})\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right) v_{p+}\left(\hat{\boldsymbol{x}},\theta_0^T(\boldsymbol{x})\right)\mathcal{T}_{p,p_+}(\theta_0(\boldsymbol{x}))\frac{e^{-\mu_p\|\boldsymbol{x}-\boldsymbol{x}_0\|}}{2\pi\|\boldsymbol{x}-\boldsymbol{x}_0\|}|\boldsymbol{J}_{\Gamma_{i_p,p+}}|\,\mathrm{d}\hat{\boldsymbol{x}}, \end{aligned} \tag{22}$$

where $\theta_0^T(\boldsymbol{x})$ is the transmitted direction in E_{p+} obtained from Snell's Law and the minus sign arises from moving the known source expression onto the right side as before. Note that there is no sum over edges here since we assume that $\Gamma_{i,p}$ coincides with a unique edge $\Gamma_{i_p,p+}$ of E_{p+}. However, a term of the form (22) will need to be computed for each element that is connected to E_p along one of its edges $\Gamma_{i,p}$, $i=1,2,\ldots,\kappa_p$. For any edges of E_p that are located along the boundary Γ, the contribution from (22) is zero. The corresponding contribution from (21) will have $\mathcal{R}_{p,p_+}(\theta) = \mathcal{R}_{p,0}(\theta) = 1$, representing conservation of energy by a specular reflection.

For the DO method, the formulation of the boundary flux basis coefficients for a source point within E_p is similar. For example, the coefficients of the source expansion (18) for the transmitted component within a homogeneous region of Ω are given by

$$u^n_{0_{p+,\beta_1,\beta_2}} = -\int_{\Gamma_{i_p,p+}} \left(\hat{\boldsymbol{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right) w_{p+}(\hat{\boldsymbol{x}}) \frac{e^{-\mu_p\|\boldsymbol{x}-\boldsymbol{x}_0\|}}{2\pi\|\boldsymbol{x}-\boldsymbol{x}_0\|}\delta\left(\theta_n-\theta_0(\boldsymbol{x})\right)\mathrm{d}\boldsymbol{x}, \tag{23}$$

noting that in this case $\mathcal{T}_{p,p_+}(\theta) = 1$ and $\theta_0^T(\boldsymbol{x}) = \theta_0(\boldsymbol{x})$. We now use the composition property of the Dirac delta to evaluate the spatial integral without changing the integration variable via F_{p+}^{-1} and Eq. (23) simplifies to

$$u^n_{0_{p+,\beta_1,\beta_2}} = \frac{w_{p+}(\hat{\boldsymbol{x}}_n)e^{-\mu_p\|\boldsymbol{x}_n-\boldsymbol{x}_0\|}}{2\pi}, \tag{24}$$

where $\boldsymbol{x}_n$ is the point where the straight line ray originating from $\boldsymbol{x}_0$ in direction θ_n intersects $\Gamma_{i_p,p+}$ and $\hat{\boldsymbol{x}}_n = F_{p+}(\boldsymbol{x}_n)$. See Appendix A for more details of the derivation of (24), and for the corresponding results in the cases of reflection and refractive transmission.

3.4. Linear system formulation and fluence evaluation

The Fourier formulation (12) and the DO scheme (14) can both be expressed compactly as a linear system of the form

$$\boldsymbol{A}\boldsymbol{u} = \boldsymbol{v}, \tag{25}$$

where

$$\boldsymbol{u} = \left(\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{N_h}\right)^T \tag{26}$$

is the global solution vector, with each $\boldsymbol{u}_j$, $j = 1, \ldots, N_h$ containing the unknown expansion coefficients $u^n_{j,\alpha_1,\alpha_2}$ from either (8) for the Fourier approach, or (10) for the DO method. The length of each $\boldsymbol{u}_j$ block is $\frac{1}{2}(k+1)(k+2)(2N+1)$ for the Fourier formulation and $\frac{1}{2}(k+1)(k+2)N_\theta$ for the DO method. The matrix $\boldsymbol{A}$ contains the coefficients of these expansion coefficients obtained from the corresponding weak Galerkin formulation, (12) or (14), for the Fourier or DO methods, respectively. The entries of $\boldsymbol{v}$ are determined from the source expansion coefficients in (17) or (18), which depend on both the choice of discretisation (Fourier or DO) and the choice of source type (boundary source or internal point source) as detailed in the previous subsection.

Once $\boldsymbol{u}$ has been obtained from (25), we can find the fluence from Eq. (3). For the Fourier method we substitute (8) into (3) to give the fluence at a point $\boldsymbol{x} \in E_j$ as

$$\boldsymbol{\Phi}_j(\boldsymbol{x}) = \sum_{\alpha_1=0}^{k}\sum_{\alpha_2=0}^{k-\alpha_1}\sum_{n=-N}^{N} u^n_{j,\alpha_1,\alpha_2}\tilde{P}_{\alpha_1}(\hat{x})\,\tilde{P}_{\alpha_2}(\hat{y})\int_{-\pi}^{\pi}\frac{e^{in\theta}}{\sqrt{2\pi}}\mathrm{d}\theta, \tag{27}$$

which can be simplified using Fourier orthogonality to give

$$\boldsymbol{\Phi}_j(\boldsymbol{x}) = \sqrt{2\pi}\sum_{\alpha_1=0}^{k}\sum_{\alpha_2=0}^{k-\alpha_1} u^0_{j,\alpha_1,\alpha_2}\tilde{P}_{\alpha_1}(\hat{x})\,\tilde{P}_{\alpha_2}(\hat{y}). \tag{28}$$

For the DO method, we similarly substitute Eq. (10) into Eq. (3) and use the Riemann-sum approximation to the angular integral. Thus in element j we obtain

$$\boldsymbol{\Phi}_j(\boldsymbol{x}) = \frac{2\pi}{N_\theta}\sum_{n=1}^{N_\theta}\sum_{\alpha_1=0}^{k}\sum_{\alpha_2=0}^{k-\alpha_1} u^n_{j,\alpha_1,\alpha_2}\tilde{P}_{\alpha_1}(\hat{x})\,\tilde{P}_{\alpha_2}(\hat{y}) \tag{29}$$

for the fluence at any point $\boldsymbol{x} \in E_j$.

For the case of an internal point source located at $\boldsymbol{x}_0 \in E_p$, we supplement the fluence $\boldsymbol{\Phi}_p$, detailed above, with the direct contribution from the source within E_p before it has reached the element boundary. This direct contribution $\boldsymbol{\Phi}_0$ corresponds to the free-space Green's function within E_p describing propagation from $\boldsymbol{x}_0$ uniformly in all directions and is given by

$$\boldsymbol{\Phi}_0(\boldsymbol{x}) = \frac{e^{-\mu_p\|\boldsymbol{x}-\boldsymbol{x}_0\|}}{2\pi\|\boldsymbol{x}-\boldsymbol{x}_0\|}.$$

Hence, for a point source, the total fluence at $\boldsymbol{x} \in E_p$ is given by $\boldsymbol{\Phi}_0(\boldsymbol{x}) + \boldsymbol{\Phi}_p(\boldsymbol{x})$, and in all other elements E_j, $j \neq p$, the fluence is simply $\boldsymbol{\Phi}_j(\boldsymbol{x})$.

4. Numerical results

In this section we apply the DG discretisation of the RTE, together with the DO and Fourier–Galerkin directional approximations introduced above, to a series of ray propagation problems. The examples are chosen to illustrate three complementary aspects of the method: accuracy and convergence on simple geometries, the treatment of internal point excitation and interfaces between different media, and the applicability of the approach to a large-scale, more complex example. All simulations were implemented in Python and performed on a 13th Gen Intel Core i7-13700 processor (16 cores, 32 GB RAM), using the BiConjugate Gradient Stabilised (BiCGStab) method to solve the linear system in Eq. (25) with a block-diagonal incomplete LU preconditioner. The matrix assembly used process-based parallelism, while the block-diagonal incomplete LU preconditioner used shared-memory threads.

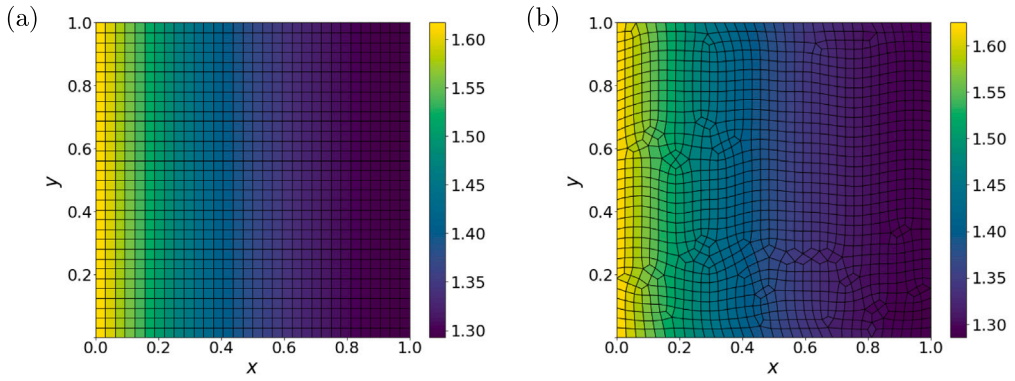


Fig. 3. Exact solution (30) for Example 1 showing the fluence generated by a boundary source propagating perpendicularly to the left edge of a square plate. The two sub-plots depict the solution on different mesh types, with plot (a) showing a regular Cartesian grid of $N_h = 1024$ square elements with $h_x = h_y = 0.03125$ and plot (b) showing a quadrilateral CALFEM [41] mesh with $N_h = 1185$ elements generated using a mesh size parameter of $h = 0.03125$.

Table 1
Error estimates (31) and estimated orders of convergence (EoC) for the fluence computed using DG-DO on regular Cartesian grids with reflective boundaries, for different numbers of elements N_h and polynomial degrees k.

N_h	$k=0$		$k=1$		$k=2$		$k=3$	
	Error	EoC	Error	EoC	Error	EoC	Error	EoC
4	1.622e−02	–	5.309e−03	–	1.741e−04	–	3.727e−06	–
16	1.019e−02	0.67	1.325e−03	2.00	2.267e−05	2.94	2.292e−07	4.02
64	5.620e−03	0.86	3.311e−04	2.00	2.891e−06	2.97	1.421e−08	4.01
256	2.940e−03	0.94	8.276e−05	2.00	3.651e−07	2.99	8.841e−10	4.01
1024	1.502e−03	0.97	2.069e−05	2.00	4.587e−08	2.99	5.514e−11	4.00
4096	7.588e−04	0.98	5.172e−06	2.00	5.748e−09	3.00	3.425e−12	4.01

4.1. Example 1: Simple aluminium plate vibrations

We begin by considering a simple benchmark problem on a square aluminium plate of thickness 4 mm, excited at a frequency of $f = \omega/(2\pi) = 20\,\text{kHz}$. The material parameters are taken to be Young's modulus: 68.9 GPa, density: $2700\,\text{kgm}^{-3}$, Poisson ratio: 0.33, and we apply a hysteretic damping loss factor of $\eta_d = 0.01$, which equates to a damping factor of $\mu = \omega\eta_d/(2c) = 0.7130303$, where the wave speed c of the flexural plate wave may be calculated from the material parameters using the standard formulæ. This example is used to verify the spatial convergence properties of the DG discretisation in a setting where an exact solution is available. The spatial domain is the unit square $\Omega = (0,1)\times(0,1)$, discretised using either a regular Cartesian grid obtained by simply making identical equispaced subdivisions along the x and y axes to generate a mesh of square elements as shown in Fig. 3(a), or using a general quadrilateral element mesh generated using the CALFEM toolbox for Python [41] as shown in Fig. 3(b). All external boundaries are taken to be perfectly reflecting, so that the specular boundary condition (2) is imposed along Γ. Excitation is introduced via a boundary source on the left edge, $\Gamma_{\text{L}} = \{0\}\times(0,1)$, in a fixed direction $\theta_0 = 0$ through the function u_0 in (2). The damping coefficient μ is taken to be constant throughout the domain, so that the resulting energy distribution is effectively one-dimensional in the x co-ordinate. For a cavity of length X, the exact 1D fluence is

$$\Phi_{\text{sol}}(x) = \frac{\cosh\big(\mu(x-X)\big)}{\sinh(\mu X)}, \qquad 0 \le x \le X, \tag{30}$$

which has been represented within both mesh types shown in Fig. 3 with $X = 1$ and $\mu = 0.7130303$, as detailed above.

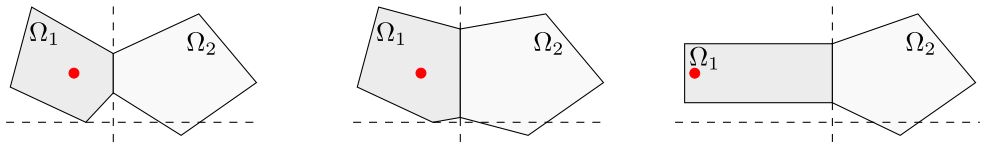
The directional dependence is discretised using $N_\theta = 2$ ordinates where $\theta_1 = -\pi$ and $\theta_2 = 0$, which therefore contains the two directions of propagation exactly. Note that the necessity for only two angular directions to represent all solution directions is particularly favourable for the DO approach, and allows us to focus solely on the approximation error arising from the spatial DG discretisation. For the spatial discretisation, we employ the DG formulation described in Section 3 with polynomial degree $k \in \{0,1,2,3\}$ on each quadrilateral element. For a given mesh and polynomial order, the numerical solution is post-processed to obtain the fluence $\boldsymbol{\Phi}_j$ for $j = 1,2,\ldots,N_h$ via (29). To quantify the error, we use an estimate of the L^2 relative error given by

$$\text{Error} = \frac{\sqrt{\sum_{j=1}^{N_h} |E_j|(\boldsymbol{\Phi}_j(\boldsymbol{x}_j) - \boldsymbol{\Phi}_{\text{sol}}(x_j))^2}}{\sqrt{\sum_{j=1}^{N_h} |E_j|(\boldsymbol{\Phi}_{\text{sol}}(x_j))^2}}, \tag{31}$$

where $|E_j|$ has been used to represent the area of the element E_j and the evaluation point $\boldsymbol{x}_j$ is taken to be the centroid of E_j. Note that x_j is simply the first component of the position vector $\boldsymbol{x}_j$. Tables 1 and 2 report these L^2 relative error estimates in the fluence

Table 2
Error estimates (31) and estimated orders of convergence (EoC) for the fluence computed using DG–DO on CALFEM quadrilateral element meshes [41] with reflective boundaries, for different numbers of elements N_h and polynomial degrees k.

N_h	$k=0$		$k=1$		$k=2$		$k=3$	
	Error	EoC	Error	EoC	Error	EoC	Error	EoC
6	3.279e−02	–	4.289e−03	–	1.115e−04	–	3.189e−06	–
21	2.080e−02	0.73	1.302e−03	1.90	2.326e−05	2.50	3.424e−07	3.56
78	6.889e−03	1.68	2.611e−04	2.45	1.791e−06	3.91	1.266e−08	5.03
299	5.280e−03	0.40	8.194e−05	1.73	3.504e−07	2.43	1.440e−09	3.24
1185	2.072e−03	1.36	1.752e−05	2.24	3.559e−08	3.32	5.134e−11	4.84
4727	9.307e−04	1.16	4.129e−06	2.09	3.948e−09	3.18	2.444e−12	4.40



(a) Configuration A (b) Configuration B (c) Configuration C

Fig. 4. Diagram showing the three coupled acoustic cavity configurations studied in [42]. In each case, the cavities are denoted Ω_j for $j = 1, 2$ and the interface between the cavities is located along the y-axis. Each system is excited by a point source located in Ω_1, which is represented by a red spot in the diagram.

and the corresponding estimated orders of convergence (EoC) for a sequence of regular Cartesian grids and a sequence of CALFEM meshes, respectively, with polynomial degrees $k = 0, 1, 2, 3$ in both cases. In each case, the mesh sequence corresponds to halving the element diameter in some sense. For the regular Cartesian grid the refinement simply corresponds to halving $h = h_x = h_y$ in each row of the table, starting with $h = 0.5$ in the first row. For the CALFEM mesh sequence, we take the same value of h used for the Cartesian grid as the mesh diameter input parameter for the CALFEM mesh generator. In both cases, the EoCs in rows $r = 2, 3, 4, 5, 6$ of the table are calculated via

$$\text{EoC} = \log_2\left(\frac{\text{Error for } h = 2^{-(r-1)}}{\text{Error for } h = 2^{-r}}\right).$$

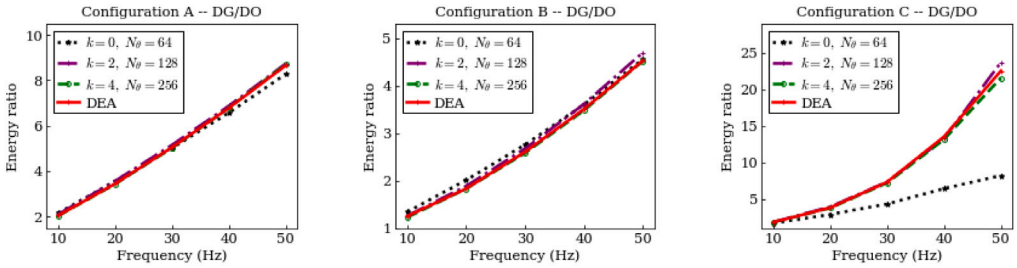
Across both Tables 1 and 2, the relative errors (31) decrease systematically under refinement, and the observed convergence rates exhibit the optimal $O(h^{k+1})$ behaviour for DG, provable for similar models under certain regularity restrictions on the mesh type [38,39]; see Appendix B for a more detailed discussion of this point. On the regular Cartesian mesh sequence reported in Table 1, the EoC is very stable, as for $k = 0$ it approaches first order, while for $k = 1, 2, 3$ the EoCs are second, third, and fourth order, respectively, over several refinement stages. In contrast, the CALFEM results in Table 2 exhibit less regular EoCs, with noticeable oscillations between successive refinements. This is not surprising since the refinement of the CALFEM quadrilateral meshes is not uniform; the elements do not all have the same area, for example. Some elements of the CALFEM mesh also violate the unique outflow condition (1.3a) in Ref. [39], which is necessary for their optimal convergence proof. However, we still recover the optimal asymptotic behaviour once the mesh is sufficiently fine.

Standard DEA treatments based on orthogonal polynomial expansions in the direction variable struggle to attain first-order convergence for this benchmark example — see Table 4.1 in Ref. [21]. However, the DO-type DEA approach introduced in [21] can easily achieve errors close to machine precision — see also Table 4.1 in Ref. [21]. On the other hand, the DO version of DEA is typically inefficient for the point source examples considered in the next section, requiring thousands of direction ordinates for a reasonable level of convergence — see Figs. 4.4 and 4.5 in Ref. [21]. The DG-DO approach presented here reproduces the DEA benchmark solutions with $N_\theta \leq 256$ as we shall see in the next section. Overall, this benchmark example confirms that our DG implementation exhibits optimal convergence rates on general quadrilateral meshes with reflective boundaries and provides an accurate approximation of the fluence for simple boundary source problems.

4.2. Example 2: Coupled acoustic cavity benchmark test

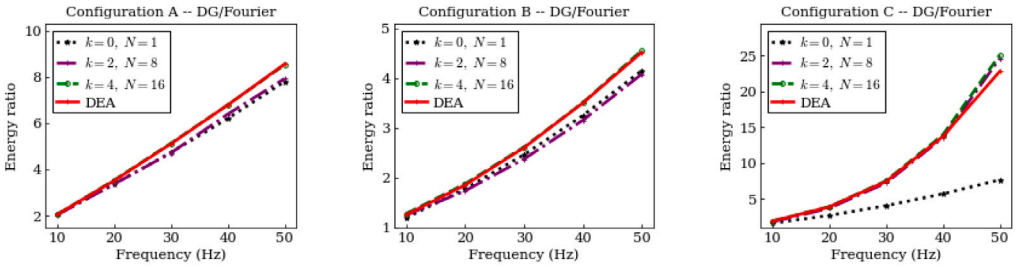
As a second benchmark, we consider the three coupled cavity configurations studied in [42], where they were used to compare DEA with SEA and finite element computations. In each configuration, the problem consists of two coupled acoustic cavities with a common interface at $x = 0$ and a (hysteretic) damping factor $\eta_d = 0.01$, corresponding to the choice $\mu_j = \omega\eta_d/(2c_j)$ within the cavity Ω_j for $j = 1, 2$. The three configurations, labelled A–C, are shown in Fig. 4; for more details see Ref. [42].

Configuration A consists of two irregular pentagonal cavities with a relatively narrow interface, giving rise to distinct energy levels in each cavity and, for sufficiently low damping, an even spreading of the acoustic energy density throughout each cavity. These factors suggest that an SEA model would likely provide reasonable results for this case. Configurations B and C are less suitable for SEA owing to the longer interface present in configuration B and the regular (wave-guiding) geometry of Ω_1 in configuration C.



(a) Configuration A (b) Configuration B (c) Configuration C

Fig. 5. Energy ratio $R_e = P_1/P_2$ for coupled-cavity configurations A–C. The plots compare the DG-DO results for $(k, N_\theta) = (0, 64), (2, 128), (4, 256)$ with the reference DEA results from [42]. The results were produced with only $N_h = 2$ elements, one per cavity, and with decay parameter $\mu_j = 0.01\omega/(2c_j)$, $j = 1, 2$ where $c_1 = c_2 = 1$.



(a) Configuration A (b) Configuration B (c) Configuration C

Fig. 6. Energy ratio $R_e = P_1/P_2$ for coupled-cavity configurations A–C. The plots compare the DG-Fourier results for $(k, N) = (0, 1), (2, 8), (4, 16)$ with the reference DEA results from [42]. The results were produced with only $N_h = 2$ elements, one per cavity, and with decay parameter $\mu_j = 0.01\omega/(2c_j)$, $j = 1, 2$ where $c_1 = c_2 = 1$.

The quantity of interest in Ref. [42] is the ratio R_e of the total energy in each cavity, calculated via

$$R_e = \frac{P_1}{P_2}, \qquad \text{where } P_j = \int_{\Omega_j} \Phi_j(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x}, \qquad j = 1, 2.$$

For each configuration shown in Fig. 4, we use a spatial discretisation with only two elements, $E_1 = \Omega_1$ and $E_2 = \Omega_2$, and compute R_e over a range of frequencies. The integrals required to evaluate P_j for $j = 1, 2$ are estimated using a midpoint quadrature rule on a triangulation of Ω_j. We apply specular reflection boundary conditions on the non-interface edges of Ω_2, that is Eq. (2) with $u_0 \equiv 0$. A source point $\boldsymbol{x}_0$ is located in Ω_1 and so on the non-interface edges of Ω_1, we apply the boundary condition (2) with u_0 corresponding to the reflected point source contributions outlined in Section 3.3 and Appendix A (see Eqs. (21) and (A.6)) with a reflection coefficient of 1. On the interface between Ω_1 and Ω_2, we apply reflection/transmission laws as outlined in Section 3.2, including in the point source contribution described in Section 3.3. Initially, we set the wave speed in both elements to be equal with $c_1 = c_2 = 1$ and throughout, we set the fluid density in both elements to be $\rho_1 = \rho_2 = 1$.

Fig. 5 presents the DG-DO results compared with the DEA reference data from [42], and Fig. 6 presents the DG-Fourier results. Across configurations A to C, the DG curves in Figs. 5 and 6 converge towards the DEA reference curve as the polynomial degree is increased and the directional discretisation is refined. In all cases, the energy ratio R_e increases with frequency, and the higher-resolution results are in excellent agreement with the DEA results across the full range of examples and frequencies. For configurations A and B, the curves are tightly grouped, and so even the lower-resolution discretisations perform reasonably well here. This is clearest in configuration A, where the lowest-order curves remain close to DEA over most of the frequency band. We note that the DG-Fourier discretisation with $k = 0$ and $N = 1$ corresponds to the SEA assumptions of local equilibrium and diffusive transport, and so it is perhaps not surprising that this case provides reasonable results for configuration A. For a full SEA treatment, it would be necessary to additionally implement diffuse reflection boundary conditions as in Ref. [29], although the impact of this would likely be minimal (and could be investigated as a future study) since, after leaving the boundary, the transport is instantaneously diffusive with $N = 1$. We also observe the success of low order DG-DO simulations ($k = 0$, $N_\theta = 64$) for configuration B.

Configuration C is more demanding owing to the directional channelling of energy along the elongated cavity Ω_1, from the source at the far left to the interface at the right. This gives rise to exponential energy decay along the channel, with the decay rate increasing in proportion to the frequency. The DEA curve rises sharply at higher frequencies (and thus higher damping) and the

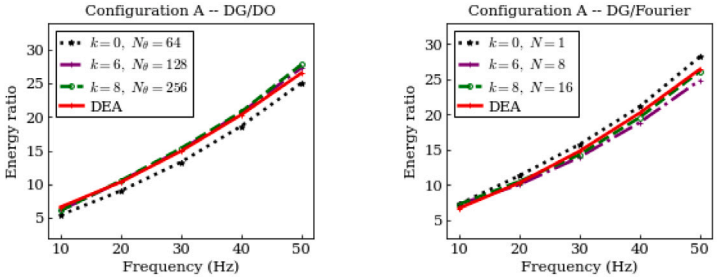


Fig. 7. Energy ratio $R_e = P_1/P_2$ for the coupled-cavity configuration A. The plots compare the DG-DO results (left) and the DG-Fourier results (right) with the reference DEA result from [42]. The results were produced with only $N_h = 2$ pentagonal elements, one per cavity, and with decay parameter $\mu_j = 0.01\omega/(2c_j)$, $j = 1, 2$ where $c_1 = 0.5$ and $c_2 = 1$.

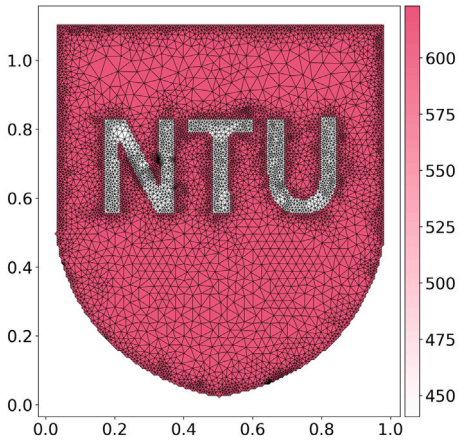
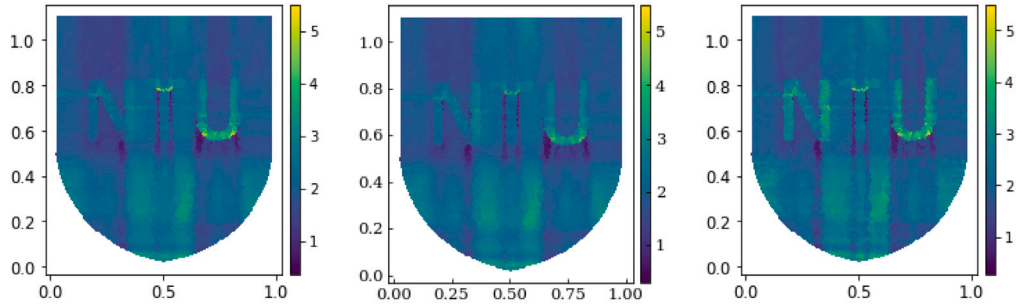


Fig. 8. NTU shield geometry represented using an unstructured triangular CALFEM mesh. The thinner NTU sub-region of thickness 2 mm represents an engraving of the thicker 4 mm surrounding shield region. The values on the colour-bar correspond to the local flexural wave speed at frequency $f = 10$ kHz.

$k = 0$ results are highly inaccurate here. Increasing the polynomial degree and refining the directional discretisation demonstrates the convergence of both (the DO and Fourier) approaches towards the DEA reference result.

We now consider a modified version of configuration A with a discontinuous wave speed across the interface, so that $c_1 = 0.5$ and $c_2 = 1$. As before, we can compare our approaches to the corresponding DEA results presented in Ref. [42]. For this example, we need to account for the reflection and transmission behaviour at the interface as described in Section 3. In Fig. 7, we plot the results from our DG-DO and DG-Fourier methods using various combinations of k, N and N_θ. Fig. 7 shows that in this case, convergence towards the DEA reference curve was slower and thus we revised the study to use higher polynomial degrees $k = 0, 6, 8$ in the DG discretisation. We find that both approaches reach excellent agreement with the DEA curve for the $k = 8$ case and that the $k = 6$ result for DG-DO also matches well.

Overall, both the DG-DO and DG-Fourier approaches converge towards the reference DEA results, agreeing well for a suitably refined discretisation. Therefore computational efficiency, as opposed to capability, becomes an important distinguishing feature. For context, the reference DEA results presented in Ref. [42] were computed using a Chebyshev polynomial basis of the same degree along each edge of each polygon, as well as in the angular direction variable. The results here are presented for sixth degree Chebyshev polynomials, which for configuration A (or B) means a total of 490 degrees of freedom with a computation time of approximately 3910 s for the serial runs reported in Table 1 of Ref. [43]. The typical computational time in seconds for the DO approach here ranges from 0.5 s for $k = 0$, $N_\theta = 64$, to 2 s for $k = 2$, $N_\theta = 128$, and 13 s for $k = 4$, $N_\theta = 256$. For the Fourier formulation, the corresponding run times are 6 s for $k = 0$, $N = 1$, 14 s for $k = 2$, $N = 8$, and 92 s for $k = 4$, $N = 16$. The Fourier simulations therefore require significantly larger computational times, and the additional computational expense does not produce a superior prediction for R_e. The additional cost of the DG-Fourier approach can be attributed to the overhead for the additional integration over direction as well as the implementation of basis/test functions in direction, which is not necessary in DO. Note also that doubling N requires $2N$ additional Fourier terms, whereas doubling N_θ requires only N_θ additional ordinates. In summary, we find that the DG-DO approach provides the more efficient route to a converged result.



(a) $k = 1,\ N_\theta = 256$ (b) $k = 1,\ N_\theta = 512$ (c) $k = 2,\ N_\theta = 256$

Fig. 9. Vibrational energy distribution on the NTU shield for DG-DO discretisations with different spatial polynomial degrees $k = 1, 2$ and different numbers of ordinates $N_\theta = 256, 512$. The excitation frequency was taken as $f = 10$ kHz.

4.3. Example 3: High-frequency vibrations of an engraved shield

The final example is intended to assess how well the DG-DO approach scales to geometries and meshes of a size and complexity comparable to those found in industrial applications. We consider an engraved aluminium plate model of the Nottingham Trent University shield logo, comprising a main shield region of thickness $h_{\text{shield}} = 4$ mm and an engraved NTU-shaped sub-region of reduced thickness $h_{\text{NTU}} = 2$ mm, resulting in an abrupt change in wave speed, as shown in Fig. 8. The material parameters for aluminium are taken to be the same as in Example 1 (Young's modulus: 68.9 GPa, density: 2700 kgm^{-3}, Poisson ratio: 0.33 and hysteretic damping loss factor $\eta_d = 0.01$). This example therefore presents several challenges including regions of differing wave speed, more intricate geometrical features than before and the approximation of curved regions by straight-line segments.

A boundary source excitation is applied along the top edge of the shield with $\theta_0 = -\pi/2$ in (19), so that energy is injected downwards into the shield interior. All external boundaries are taken to be specularly reflecting. The spatial discretisation is performed on an unstructured triangular mesh with 12 822 elements, as shown in Fig. 8. The total number of degrees of freedom is

$$N_{\text{dof}} = N_h \frac{1}{2}(k+1)(k+2)N_\theta, \tag{32}$$

which for $N_h = 12\,822$, $k = 1$ and $N_\theta = 256$ yields $N_{\text{dof}} \approx 9.8 \times 10^6$, increasing to roughly 2.0×10^7 for either $k = 1$, $N_\theta = 512$ or $k = 2$, $N_\theta = 256$. This problem therefore provides a test of the scalability of the method to industry-scale problems with millions of degrees of freedom. In the absence of an exact solution, we perform a convergence study in Fig. 9, taking a higher resolution result as a reference solution.

Fig. 9 shows the computed fluence distribution on the shield for three discretisation choices at a frequency of 10 kHz and using the reflection/transmission coefficients derived from the four equation model including evanescent waves discussed in Section 3.2. In all three cases, the main qualitative features are the same. The energy enters through the top boundary and propagates downwards, forming a vertically banded pattern with shadowing below the engraved letters due to the reflections at the interfaces where the thickness changes. The thinner engraved region is also clearly visible owing to this change in thickness. Comparing the three panels, only very minor changes in the fluence are observed when either the angular resolution is increased from $N_\theta = 256$ to $N_\theta = 512$, or the spatial polynomial degree is increased from $k = 1$ to $k = 2$. The overall energy distribution and the energy density levels in both the main shield and the engraved region are in excellent agreement, indicating that a baseline choice of $k = 1$, $N_\theta = 256$ already provides a reasonably well converged solution on this mesh.

We next assess the interface reflection/transmission coefficient model by comparing two alternative formulations of the heterogeneous interface conditions, using the baseline discretisation $k = 1$, $N_\theta = 256$ at frequencies of 5 kHz, 10 kHz and 20 kHz. In particular, we compare the four-equation model including evanescent contributions, as detailed in Section 3, with a reduced two-equation approximation obtained by neglecting the evanescent components. The latter leads to the reflection and transmission coefficients stated in Eq. (16). Fig. 10 shows the vibrational energy distributions for these three frequencies, comparing the two- and four-equation formulations.

At 5 kHz there are noticeable, though still modest, differences between the fluence obtained from the two models in the vicinity of the internal interfaces as the four-equation formulation predicts a slightly stronger localisation of energy near the thickness transition, demonstrating the influence of the evanescent components. As the frequency is increased to 10 kHz, and then to 20 kHz, these differences diminish. The fluence plots for the two interface formulations become very similar at both 10 kHz and 20 kHz, showing excellent agreement. We therefore observe the expected trends as the frequency is increased, meaning that we can neglect the evanescent terms and make use of the simpler two-equation model for sufficiently high frequencies.

The computational cost of this larger engraved shield calculation remains moderate. For the $k = 1$ and $N_\theta = 256$ base case, the full DG-DO work flow required approximately 221 s using 16 workers with a parallel efficiency of 28% and a peak memory requirement of 9.9 GB (with 4 workers the time is 314 s with 78% efficiency and a peak memory requirement of 8.9 GB). The sparse

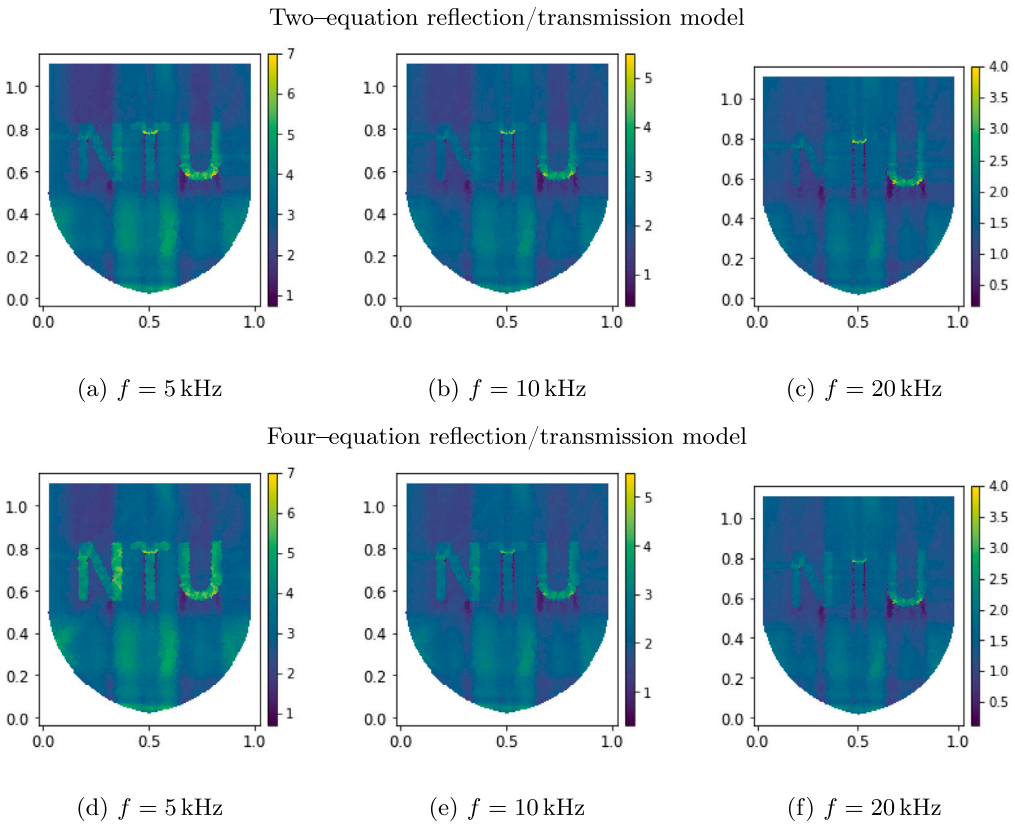


Fig. 10. Vibrational energy distributions on the NTU shield using the reduced two-equation reflection/transmission coefficient model (top row) and the full four-equation model (bottom row) at three different excitation frequencies.

matrix $\boldsymbol{A}$ appearing in the linear system (25) has dimension 9847296 × 9847296 with just under 84 million non-zero entries. Given the nearly ten million degrees of freedom involved, this run time demonstrates that the proposed formulation is capable of handling multi-millions of degrees of freedom on complex geometries within a competitive time-frame.

5. Conclusions

This paper has presented a new approach for the numerical simulation of high-frequency wave problems in the asymptotic ray-tracing limit, based on the radiative transfer equation. We have drawn upon the discontinuous Galerkin techniques commonly applied in nuclear reactor analysis and atmospheric science and revised the formulation (see Section 3.2 to 3.4) to instead model high-frequency wave simulations in multi-media or multi-material domains including wave reflection/transmission and refraction effects. We have considered both the more standard discrete ordinates formulation for approximating the directional behaviour of the wave field, as well as a Fourier–Galerkin discretisation whose low-order implementation is equivalent to the SEA approach commonly used to estimate high-frequency vibrational energy within complex structures.

We then performed a sequence of numerical tests to verify the approach and showcase its capability. In particular, Example 1 demonstrated optimality of the DG discretisation for a simple bi-directional ray flow corresponding to flexural waves on a simple aluminium plate with $\mathcal{O}(h^{k+1})$ convergence estimates. On the other hand, in Example 2 we fixed a coarse mesh of two elements to model pairs of coupled acoustic cavities, demonstrating the flexibility of the approach in terms of incorporating more general polygonal elements and wave reflection/transmission effects. We studied the convergence as the polynomial degree k is increased, comparing also the effectiveness of both the discrete ordinates and Fourier directional discretisations. We found that both approaches converge to the reference data from previous studies, but the computational efficiency of the discrete ordinates approach is superior and grows more slowly under refinement of the scheme. The Fourier approach was only competitive in the regime where the assumptions typical for the success of SEA are satisfied and a coarse implementation suffices. The final example demonstrates the capability of the discrete ordinates approach to scale up to problems including more intricate geometrical features requiring fine meshes, as well as wave reflection/transmission effects. The results for this example demonstrate the high-frequency validity of simplified reflection/transmission coefficients for flexural waves in plates, obtained by omitting the evanescent components, as well as the efficiency of the approach for computing an example with almost ten million degrees of freedom in under 4 minutes.

CRediT authorship contribution statement

Robert J. Lockett: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **David J. Chappell:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Point source contributions using the DO method

In this appendix we give the details of the point source contributions in the DG-DO scheme. We initially detail the evaluation of the integral in (23), before generalising to element boundaries where the neighbouring elements have different wave properties as well as the reflected contribution in E_p.

Transmitted component in a homogeneous region

Firstly we consider the application of the composition property of the Dirac delta by expressing the integral (23) as

$$u^n_{0_{p+,\beta_1,\beta_2}} = -\int_{\Gamma_{i_p,p+}} \left(\hat{\boldsymbol{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right) w_{p+}(\hat{\boldsymbol{x}})\frac{e^{-\mu_p\|\boldsymbol{x}-\boldsymbol{x}_0\|}}{2\pi\|\boldsymbol{x}-\boldsymbol{x}_0\|}\delta\big(g(\boldsymbol{x})\big)\mathrm{d}\boldsymbol{x} \tag{A.1}$$

with $g(\boldsymbol{x}) = \theta_n - \theta_0(\boldsymbol{x})$ and where we have removed the Jacobian term to express the integral with respect to global spatial coordinates $\boldsymbol{x}$. We note that this integral is restricted to a straight line polygonal edge and so we introduce s as an arclength parameter along the edge $\Gamma_{i_p,p+}$ over which we may express the integral (A.1). We note that

$$\left|\frac{\mathrm{d}g}{\mathrm{d}s}\right| = \left|\frac{\mathrm{d}\theta_0}{\mathrm{d}s}\right|, \tag{A.2}$$

where θ_0 can be calculated via

$$\theta_0(x,y) \;=\; \tan^{-1}\left(\frac{y-y_0}{x-x_0}\right).$$

Calculating the spatial derivatives then gives

$$\frac{\partial\theta_0}{\partial x} = -\frac{\sin\left(\theta_0\right)}{\|\boldsymbol{x}-\boldsymbol{x}_0\|}, \qquad \frac{\partial\theta_0}{\partial y} = \frac{\cos\left(\theta_0\right)}{\|\boldsymbol{x}-\boldsymbol{x}_0\|}.$$

In order to apply the composition property of the Dirac delta, the derivative (A.2) should be evaluated at the point $\boldsymbol{x}_n = (x_n, y_n)$ with corresponding arclength coordinate $s = s_n$ where the ray leaving $\boldsymbol{x}_0$ in direction θ_n meets the edge $\Gamma_{i_p,p+}$ (the integral will only be non-zero if the ray with direction θ_n intersects the edge $\Gamma_{i_p,p+}$). Considering the unit tangent vector (τ_x, τ_y) along $\Gamma_{i_p,p+}$, we have

$$\frac{\mathrm{d}x}{\mathrm{d}s} = \tau_x, \quad \frac{\mathrm{d}y}{\mathrm{d}s} = \tau_y.$$

Hence,

$$\left|\frac{\mathrm{d}\theta_0}{\mathrm{d}s}\right|_{s=s_n} = \frac{\left|-\sin\left(\theta_n\right)\,\tau_x + \cos\left(\theta_n\right)\tau_y\right|}{\|\boldsymbol{x}_n-\boldsymbol{x}_0\|} = \frac{\left|\hat{\boldsymbol{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}_n)\right|}{\|\boldsymbol{x}_n-\boldsymbol{x}_0\|},$$

where we note that the outward normal vector may be expressed in terms of the components of the tangent vector via $\hat{\boldsymbol{n}}(\boldsymbol{x}_n) = \left(\tau_y, -\tau_x\right)^T$.

Combining the above, the integral (A.1) may be evaluated via

$$u^n_{0_{p+,\beta_1,\beta_2}} = -\frac{\left(\hat{\boldsymbol{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right)}{\left|\frac{\mathrm{d}\theta_0}{\mathrm{d}s}\right|_{s=s_n}} w_{p+}(\hat{\boldsymbol{x}}_n)\frac{e^{-\mu_p\|\boldsymbol{x}_n-\boldsymbol{x}_0\|}}{2\pi\|\boldsymbol{x}_n-\boldsymbol{x}_0\|} = w_{p+}(\hat{\boldsymbol{x}}_n)\frac{e^{-\mu_p\|\boldsymbol{x}_n-\boldsymbol{x}_0\|}}{2\pi} \tag{A.3}$$

as stated in (24). Note that the minus sign vanishes since $\hat{\boldsymbol{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}) < 0$.

Transmitted component in an inhomogeneous region

In the case where the material/medium properties differ between E_p and one of its neighbours E_{p+}, then the resulting source contribution cannot be simplified to the same extent since refraction will take place meaning that

$$\theta_0^T(\boldsymbol{x}) \neq \theta_0(\boldsymbol{x}).$$

The integral (A.1) is therefore modified by instead setting $g(\boldsymbol{x}) = \theta_n - \theta_0^T(\boldsymbol{x})$, which may be treated in a similar way to the previous case noting that the local transmission angle $\theta_{0_T}^l \in (-\pi/2, \pi/2)$ is related to the global angle $\theta_0^T(\boldsymbol{x}) \in [-\pi, \pi)$ via a change of orientation plus a translation by a local constant (along each edge) given by the angle between the positive x-axis and $\hat{\boldsymbol{n}}(\boldsymbol{x})$. Crucially, for a source contribution along the fixed polygonal edge $\Gamma_{i_p,p+}$ we have

$$\left|\frac{\mathrm{d}g}{\mathrm{d}s}\right| = \left|\frac{\mathrm{d}\theta_0^T}{\mathrm{d}s}\right| = \left|\frac{\mathrm{d}\theta_{0_T}^l}{\mathrm{d}s}\right|. \tag{A.4}$$

Using Snell's Law we obtain

$$\theta_{0_T}^l(\boldsymbol{x}) = \sin^{-1}\left(\frac{c_{j+}}{c_j}\sin(\theta_0^l(\boldsymbol{x}))\right) = \sin^{-1}\left(\frac{c_{j+}}{c_j}(\tau_x\cos(\theta_0(\boldsymbol{x})) + \tau_y\sin(\theta_0(\boldsymbol{x})))\right),$$

where θ_0^l is θ_0 in local incident angle coordinates. Calculating the spatial derivatives as before leads to

$$\frac{\partial\theta_{0_T}^l}{\partial x} = \frac{c_{j+}(\tau_y\cos(\theta_0(\boldsymbol{x})) - \tau_x\sin(\theta_0(\boldsymbol{x})))}{\sqrt{c_j^2 - c_{j+}^2\sin^2\left(\theta_0^l(\boldsymbol{x}_n)\right)}}\frac{\partial\theta_0}{\partial x},$$

with an analogous result for the partial derivative with respect to y. Combining these results as in the locally homogeneous region case leads to

$$\left|\frac{\mathrm{d}\theta_{0_T}^l}{\mathrm{d}s}\right|_{s=s_n} = \frac{c_{j+}(\hat{\boldsymbol{s}}\left(\theta_0\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}))}{\sqrt{c_j^2 - c_{j+}^2\sin^2\left(\theta_0^l(\boldsymbol{x}_n)\right)}}\left|\frac{\mathrm{d}\theta_0}{\mathrm{d}s}\right|_{s=s_n}.$$

Now substituting to evaluate the integral (A.1) and making use of the result from the previous section that

$$\left|\frac{\mathrm{d}\theta_0}{\mathrm{d}s}\right|_{s=s_n} = \frac{\left|\hat{\boldsymbol{s}}\left(\theta_0\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}_n)\right|}{\|\boldsymbol{x}_n - \boldsymbol{x}_0\|},$$

we obtain

$$u^n_{0_{p+,\beta_1,\beta_2}} = -w_{p+}(\hat{\boldsymbol{x}}_n)\frac{e^{-\mu_p\|\boldsymbol{x}_n-\boldsymbol{x}_0\|}}{2\pi}\frac{(\hat{\boldsymbol{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}))\sqrt{c_j^2 - c_{j+}^2\sin^2\left(\theta_0^l(\boldsymbol{x}_n)\right)}}{c_{j+}(\hat{\boldsymbol{s}}\left(\theta_0\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}))^2}\mathcal{T}_{p,p+}(\theta_0(\boldsymbol{x}_n)).$$

Reflected component

Finally, we consider the reflected contribution in E_p given by

$$u^n_{0_{p,\beta_1,\beta_2}} = -\sum_{i=1}^{\kappa_p}\int_{\Gamma_{i,p}}\left(\hat{\boldsymbol{s}}\left(\theta_n\right)\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\right)w_p(\hat{\boldsymbol{x}})\frac{e^{-\mu_p\|\boldsymbol{x}-\boldsymbol{x}_0\|}}{2\pi\|\boldsymbol{x}-\boldsymbol{x}_0\|}\delta\big(g(\boldsymbol{x})\big)\mathcal{R}_{p,p+}(\theta_0(\boldsymbol{x}))\mathrm{d}\boldsymbol{x} \tag{A.5}$$

with $g(\boldsymbol{x}) = \theta_n - \theta_0^R(\boldsymbol{x})$. We apply the same techniques as we did for the transmitted contributions within a homogeneous region and note that the integral will only be non-zero along one of the edges $\Gamma_{i,p}$, $i = 1, 2, \ldots, \kappa_p$, where the ray leaving $\boldsymbol{x}_0$ in direction θ_n intersects the boundary of E_p (intersections between source rays and vertices are avoided). The integral for the reflected contribution therefore simplifies to

$$u^n_{0_{p,\beta_1,\beta_2}} = w_p(\hat{\boldsymbol{x}}_n)\frac{e^{-\mu_p\|\boldsymbol{x}_n-\boldsymbol{x}_0\|}}{2\pi}\mathcal{R}_{p,p+}(\theta_0(\boldsymbol{x}_n)). \tag{A.6}$$

Appendix B. Optimal convergence in Example 1

In this appendix, we discuss further the optimal convergence observed in the results of Example 1, presented in Tables 1 and 2 within Section 4.1. We will restate this example problem in order to relate it to the theory in Ref. [39], and discuss/conjecture when our treatment of this example is likely (or not) to satisfy a similar theoretical underpinning.

Example 1 has constant attenuation $\mu > 0$ and only includes two propagation directions $\hat{\boldsymbol{s}} = \boldsymbol{\beta} = (1, 0)$ and $\hat{\boldsymbol{s}} = -\boldsymbol{\beta} = (-1, 0)$. As such, the problem can be reduced to a pair of equations in the spatial variable only. The model then takes a similar form to the one analysed in Ref. [39], except with a pair of equations coupled by the boundary conditions. In particular, we have

$$\boldsymbol{\beta}\cdot\nabla\phi + \mu\phi = 0 \ \text{ in } \Omega, \tag{B.1}$$

$$-\boldsymbol{\beta}\cdot\nabla\psi+\mu\psi=0 \text{ in } \Omega, \tag{B.2}$$

$$\phi=\psi+g \text{ on } \Gamma^{-}, \tag{B.3}$$

$$\psi=\phi \text{ on } \Gamma^{+}. \tag{B.4}$$

Here $\Gamma^{-}=\{\boldsymbol{x}\in\Gamma:\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})<0\}$, $\Gamma^{+}=\{\boldsymbol{x}\in\Gamma:\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})>0\}$ and the fluence is given by $\Phi=\phi+\psi$. For Example 1, we also have that $g\equiv 1$. In the notation of Section 3.1, $N_\theta=2$, $\theta_1=-\pi$, $\theta_2=0$, $u_j^1=\psi|_{E_j}$ and $u_j^2=\phi|_{E_j}$. Note that the regular Cartesian grid mesh shown in Fig. 3(a) satisfies equivalent flow conditions to those proposed in (1.3) of [39], whereas the CALFEM mesh shown in Fig. 3(b) may violate the requirement for each element to have a unique outflow face with respect to $\pm\boldsymbol{\beta}$. In what follows, we therefore limit our discussion to the regular Cartesian grid mesh. We note that every element E_j, $j=1,2,\ldots,N_h$ is identical (after translation) and consider the family of meshes $\{\Omega_h\}$ of Ω where $h=\sqrt{h_x^2+h_y^2}$ denotes the element diameter.

We now introduce the upwind DG approximation in the finite dimensional space V_h^k containing polynomials of degree at most k on each element E_j. We consider DG approximate solutions $(\phi_h,\psi_h)\in V_h^k\times V_h^k$ of (B.1)–(B.4) which satisfy

$$B_{+}(\phi_h,w_h)+<\psi_h,w_h\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}>_{\Gamma^{-}}=-<g,w_h\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}>_{\Gamma^{-}}, \tag{B.5}$$

$$B_{-}(\psi_h,w_h)-<\phi_h,w_h\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}>_{\Gamma^{+}}=0, \tag{B.6}$$

for all $w_h\in V_h^k$. Here, B_+ is the bilinear form defined in (2.1b) of Ref. [39] (noting the notational differences with [39]: $\mu\leftrightarrow c$, $\hat{\boldsymbol{n}}\leftrightarrow\boldsymbol{n}$ and $\Omega_h\leftrightarrow\mathcal{T}_h$) and B_- is the bilinear form obtained by exchanging $\boldsymbol{\beta}$ for $-\boldsymbol{\beta}$ and Γ^- for Γ^+ in B_+. We are also using the notation

$$<v,w_h\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}>_{\Gamma^{\pm}}=\sum_{j=1}^{N_h}\int_{\partial E_j\cap\Gamma^{\pm}}v(\boldsymbol{x})w_h(\boldsymbol{x})(\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}(\boldsymbol{x}))\mathrm{d}\boldsymbol{x}$$

for any $v\in H^1(\Omega_h)$.

The projection $\mathbb{P}$ introduced in (2.3) of Ref. [39] can be directly generalised to our Cartesian grid mesh and we conjecture that in doing so, one would obtain the approximation property

$$\|\mathbb{P}v-v\|_{L^2(E_j)}\leq Ch^{k+1}|v|_{H^{k+1}(E_j)} \tag{B.7}$$

with $C>0$ a constant and $v=\phi,\psi$ on the elements E_j, $j=1,2,\ldots,N_h$ of our Cartesian grid mesh. Since we are using equivalent bilinear forms (up to a sign change on some terms that will subsequently vanish) to [39], we may also follow the proof of (Thm. 2.2, [39]) having only the second terms on the left of (B.5) and (B.6) to deal with as additional terms not present in [39]. We conjecture that upon evaluating

$$<\psi-\mathbb{P}\psi,w\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}>_{\Gamma^{-}}$$

and

$$<\phi-\mathbb{P}\phi,w\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}>_{\Gamma^{+}},$$

the properties of the projection $\mathbb{P}$ could be invoked/generalised to ensure that these terms vanish for all $w\in V_h^k$. Following the remaining steps of the proof of (Thm. 2.2, [39]) and applying the approximation property (B.7) would then provide optimal convergence estimates

$$\|v-v_h\|_{L^2(\Omega_h)}\leq C_v h^{k+1}|v|_{H^{k+1}(\Omega_h)} \tag{B.8}$$

on $v=\phi,\psi\in H^{k+1}(\Omega_h)$ and with $C_v>0$ the constants for each estimate. The triangle inequality then leads to an estimate for the approximate fluence $\Phi_h=\phi_h+\psi_h$ (which is the quantity evaluated in our numerical results in Tables 1 and 2) as follows

$$\begin{aligned}\|\Phi-\Phi_h\|_{L^2(\Omega_h)}&=\|(\phi-\phi_h)+(\psi-\psi_h)\|_{L^2(\Omega_h)}\\&\leq\|\phi-\phi_h\|_{L^2(\Omega_h)}+\|\psi-\psi_h\|_{L^2(\Omega_h)}\\&\leq(C_\phi|\phi|_{H^{k+1}(\Omega_h)}+C_\psi|\psi|_{H^{k+1}(\Omega_h)})h^{k+1}.\end{aligned}$$

Data availability

Data will be made available on request.

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